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Robust Portfolio Construction and Risk Analysis

Preface

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This Robust Portfolio Construction and Risk Analysis (RPCRA) book, with its unique *reproducibility* code, *robust estimators*, and *special focus* topics, has been written for use in advanced upper level undergraduate, M.S., and PhD level quantitative finance courses. It grew out of our collective experiences as researchers, educators and investors, and is thus supportive of research projects in such programs, as well as in commercial quantitative portfolio management and research.

Much of the content in Chapters 1-9 is suitable for a full academic year M.S. course, and selected portions of these chapters can serve the needs of a one-semester course. The additional material in Chapters 10-16 is well suited to special topics and project oriented courses. The chapters and the topics they cover are:

- Chapter 1. Introduction to Portfolio Management
- Chapter 2. Foundations
- Chapter 3. Modeling Asset Returns
- Chapter 4. Covariance Matrices
- Chapter 5. Portfolio Constraints and Penalties
- Chapter 6. Time Series Models
- Chapter 7. Cross Section Factor Models
- Chapter 8. Portfolio Risk Analysis
- Chapter 9. Minimum Downside Risk Portfolios
- Chapter 10. Performance Evaluation
- Chapter 11. Expected Returns
- Chapter 12. Fixed Income Portfolio Management
- Chapter 13. Portfolios With Options
- Chapter 14. Managing Portfolios to Meet Spending Objectives
- Chapter 15. Influence Functions and Standard Errors
- Chapter 16. Robust Estimators

In addition, there are five appendices

- Appendix A. Mathematical Tools For PCRA
- Appendix B. Mathematical Tools For PCRA
- Appendix C. Utility Theory
- Appendix D. Portfolio Theory
- Appendix E. Machine Learning

Reproducibility

R packages that include R functions and data support reproducibility of figures and tables throughout the book. Specifically, this is the case for most of the figures and tables in Chapters 2-6 and 9-12, as well as the figures and tables in some sections of Chapters 1 and 7-8. These features allow students to get hands-on experience in replicating the figures and tables in RPCRA, and in extending the code to study portfolio performance for different different time periods, different portfolio stock groups (such as capitalization groups and sectors), and different returns frequencies. These features also support instructors in creating new computing-oriented portfolio construction and risk analysis exercises.

R Packages for the RPCRA Book and its Reproducibility Character

The following R packages were developed specifically for the writing of the RPCRA book and its reproducibility character, and are downloadable from the CRAN website <https://cran.r-project.org>:

- PCRA: Portfolio construction and risk analysis
- facmodCS: Cross-section factor models
- facmodTS: Time series factor models
- robustGarch: Robust Garch(1, 1) models.

Data for Reproducibility

The PCRA package contains data sets that support reproducibility as well as a multitude of exercises at the end of each chapter. They may also be used by course instructors to create new exercises. The datasets include:

- Center for Research of Security Prices (CRSP[®]) monthly, weekly and daily stocks data for 294 stocks from 1993 to 2015, and Standard and Poors Global Market Intelligence (SPGMI) monthly factors data for the CRSP[®] stocks data (for reproducibility of results in Chapters 2-9)
- S&P, CBOE, and Hedge Fund data sets for reproducibility in Chapters 10 and 11.
- ICE Fixed Income datasets for reproducibility in Chapter 12

An overview of the PCRA package and its data sets are provided in the *PCRA Package and Data Overview* vignette that is downloadable from <https://cran.r-project.org/web/packages/PCRA>. Detailed information about the CRSP[®] monthly, weekly and daily stocks data, and factorsSPGMI monthly data is provided in the *CRSP Stocks and SPGMI Factors in PCRA* vignette that is also downloadable from the above PCRA package link.

Robust Estimators

Having firsthand experience of the impact of outliers in portfolio management, we were struck by how little attention has been paid to this important problem in portfolio management textbooks, and more generally in quantitative finance books. More often than not, students learn the elegant mathematics of portfolio theory with little appreciation for its fragility in the face of the prodigious levels of noise that permeate most markets. Such noise includes outliers that can have large distorting influences on the classical estimators that are routinely used in risk analysis.

The adverse influence of outliers on these classical estimators can be avoided by using modern robust alternatives that are not much influenced by outliers, and which provide better fits to the bulk of the data than do classical estimators. Chapters 2, 4, 6, and 7 introduce modern robust estimators of returns means and standard errors, covariance matrices, time-series factor models, and cross-section factor models respectively. These chapters provide clear examples of the superior fits afforded by robust estimators relative to classical estimators in the presence of even a few outliers. Chapter 16 provides background mathematical details on robust estimators and provides a bridge to RobStat™, the Maronna et al. (2019) *Robust Statistics: Theory and Methods* book that serves as our primary reference for robust statistical methods.

The robust estimators examples in this book suggest that they may result in better portfolio performance than classical estimators. But this is not guaranteed – extensive studies are needed to determine if and when robust estimators can consistently result in better portfolio performance. That said, robust estimators serve a very useful purpose of diagnosing whether or not outliers are influencing a classical estimator model fit, for, without such comparison, portfolio managers will typically be unaware of whether or not outliers are influencing the classical estimators they typically use. This claim is vividly demonstrated when comparing robust and classical regression model fits in Chapters 6 and 7 (time series and cross-section factor models, respectively).

Special Focus Topics

Chapters 10-16 cover a number of special topics that most textbooks either ignore or address with a light touch. We hope in this book to rectify these omissions.

In Chapter 10, we cover performance measurement for public and private funds and the application of statistical process control techniques (the CUSUM algorithm) to rapidly detect underperformance for actively managed portfolios. Chapter 11 develops a powerful accounting-based framework to estimate the expected returns of equity indices and also addresses the identification of business cycles for purposes of conditional forecasting. In addition, it covers the estimation of factor risk premia and touches on the reproducibility and economic viability of the vast universe of reported anomalies. Some of the material in this chapter is well known to economists and electrical engineers, particularly those focused

on signal processing, but lacks visibility in finance. We also explore the links between expected returns, corporate finance and corporate valuation, all three of which are joined at the hip intellectually, but are usually treated separately.

Chapter 12 addresses fixed income markets and ends with a discussion of practical techniques for robust fixed income portfolio management and immunization of liabilities. Chapter 13 explores the role of options in portfolios, while Chapter 14 explores end uses of portfolio management – in particular, the management of portfolios to meet spending objectives. More often than not, these topics are treated separately, but it is impossible to think constructively about one without keeping the other in mind. Arguably, re-centering one's thoughts around an asset-liability framework instead of an asset-only framework is the single most important change that an investor can make, and we hope to nudge our readers in this direction.

Finally, in Chapters 15 and 16, we return to robust statistics, and develop ideas presented earlier in the book in greater detail.

Acknowledgments

This book, and its accompanying PCRA package, would not have seen the light of day without help from a vast array of colleagues, friends and well wishers. Some helped with the development and maintenance of one or more R packages that support the the book and its associated reproducibility character. Others read chapters and commented on them, and yet others granted us formal rights to include datasets in the PCRA package that are not available to the public. These contributors are listed in the following paragraphs; we thank them for their generosity, and apologize in advance to any we might inadvertently have missed.

First, the people who helped develop and maintain the key R packages that support this book and its reproducibility features: Xinran Zhao for `PortfolioAnalytics`, Brian Peterson for `PerformanceAnalytics`, Mahmood (Mido) Shamma for `facmodCS`, Jon Spinney for PCRA and `facmodCS`, and Yuhong Liu for `robustGarch`. Additionally, Vinav Singh Sancheti wrote the code for the CUSUM algorithm.

Next, our readers: David Carino, Roland Lochoff, George Mylnikov, Dale Rosenthal, Nikolay Ryabkov, Jon Spinney, Stoyan Stoyanov and Jarrod Wilcox read and commented on multiple chapters. Their inputs proved invaluable and led us to substantially rewrite every single chapter they read. Stoyan, in particular, is a long-time friend and $\text{LyX}/\text{\LaTeX}$ whiz who helped clarify our thinking, and without whose guidance we would have struggled to get the book ready for production. Doug and Tom had the benefit of outstanding Ph. D students and teaching assistants (Yindeng Jiang, Shengyu Zhang, Ercument Cahan, Cheng Xu and Yuchen Han) who corrected their errors with grace. It's fair to say that we learned as much from our readers, students and teaching assistants as they did from us.

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Disclaimers Related to PCRA Datasets

Thanks to the generosity of a number of industry sponsors, we have been able to supply historical and market data for a variety of securities, indices and factors. In all cases, the data provided with the PCRA package that accompanies this book is *strictly for educational (i.e. non-commercial) use* – anyone who wishes to use the data for commercial purposes must enter into an appropriate licensing agreement with the relevant vendor. Furthermore, the data cannot be redistributed or published elsewhere. Please read the relevant vendor disclosures below.

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References

Maronna, Ricardo A. et al. (2019). *Robust Statistics: Theory and Methods (with R)*. 2nd ed. John Wiley & Sons.